SQL `ROW_NUMBER()` window function

`ROW_NUMBER()` assigns a unique sequential number to each row in the result, based on the ordering you specify.

It is commonly used to find the Nth-highest salary, latest record per customer, remove duplicates, and implement pagination.

1. Basic example

Suppose we have this employee table:

Employee	Department	Salary
Amit	IT	50000
Rahul	IT	70000
Priya	IT	60000
Neha	HR	40000
Ravi	HR	50000

Query:

```
SELECT
    employee,
    department,
    salary,
    ROW_NUMBER() OVER (
        ORDER BY salary DESC
    ) AS rn
```

FROM employees;

Result:

Employee	Department	Salary	RN
Rahul	IT	70000	1
Priya	IT	60000	2
Amit	IT	50000	3
Ravi	HR	50000	4
Neha	HR	40000	5

Because we specified `ORDER BY salary DESC`, the highest salary receives row number 1.

Note that Amit and Ravi have equal salaries, so their relative order is not guaranteed. Add a unique tie-breaker, such as `employee_id`, when deterministic ordering matters.

2. `ROW_NUMBER()` with `PARTITION BY`

Now suppose we want to number employees separately within each department.

```
SELECT
   employee,
   department,
   salary,
   ROW_NUMBER() OVER (
      PARTITION BY department
      ORDER BY salary DESC
   ) AS rn
FROM employees;
```

Result:

Employee	Department	Salary	RN
Rahul	IT	70000	1
Priya	IT	60000	2
Amit	IT	50000	3
Ravi	HR	50000	1
Neha	HR	40000	2

Visual explanation

IT partition

Rahul

1

Priya

2

Amit

3

HR partition

Ravi

1

Neha

2
Each partition restarts numbering from 1.

3. Find the highest-paid employee in each department

This is a classic interview question.

```
SELECT employee, department, salary
FROM (
   SELECT
      employee,
      department,
      salary,
      ROW_NUMBER() OVER (
         PARTITION BY department
         ORDER BY salary DESC, employee ASC
      ) AS rn
   FROM employees
)
WHERE rn = 1;
```

Result:

Employee	Department	Salary
Rahul	IT	70000
Ravi	HR	50000

Here, `rn = 1` selects the first employee in each department. If multiple employees have the same highest salary, `ROW_NUMBER()` selects just one according to the specified tie-breaker.

4. Difference between `ROW_NUMBER()`, `RANK()` and `DENSE_RANK()`

Suppose salaries are `90000, 80000, 80000, 70000`.

Salary	ROW_NUMBER()	RANK()	DENSE_RANK()
90000	1	1	1
80000	2	2	2
80000	3	2	2
70000	4	4	3

- `ROW_NUMBER()`: always assigns unique numbers.
- `RANK()`: tied rows share a rank, and the next rank has a gap.
- `DENSE_RANK()`: tied rows share a rank, without gaps.

5. Important interview point

In Oracle, you generally cannot filter a window function's result in the `WHERE` clause at the same query level where it is calculated. Use a subquery or a CTE:

```
WITH ranked_employees AS (
   SELECT
      e.*,
      ROW_NUMBER() OVER (
         PARTITION BY department
         ORDER BY salary DESC
      ) AS rn
   FROM employees e
)
SELECT *
FROM ranked_employees
WHERE rn <= 2;
```

This returns the top two employees per department, with exactly two rows per department when enough employees exist.